

PHYSICS — SEMESTER - I

3 lesson(s) included, 0 missing

Included:

- Theme 1 : Matter
- Theme 2 : Physical Quantities and Measurement
- Theme 3 : Force

The Vikasa School, Tuticorin
Question Bank

Sub : Physics

Class : VI

Ch 1 : Matter

Question 1 :

Choose the best option

1. Which state of matter has a definite shape and a definite volume?
(a) Liquid (b) Gas
(c) Solid (d) Plasma
2. In which state of matter are the molecules most loosely packed?
(a) Solids (b) Liquids
(c) Gases (d) Crystals
3. Which of the following is NOT a property of a liquid?
(a) Definite volume (b) Definite shape
(c) Ability to flow (d) Slight compressibility

Assertion and Reasoning

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is NOT the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

1. **Assertion (A):** A gas fills the entire space of the container in which it is kept.
Reasoning (R): The molecules of a gas have very weak forces of attraction and move randomly in all directions.
2. **Assertion (A):** Intermolecular force of attraction is maximum in gases.
Reasoning (R): Molecules in a gas are very far apart from each other.

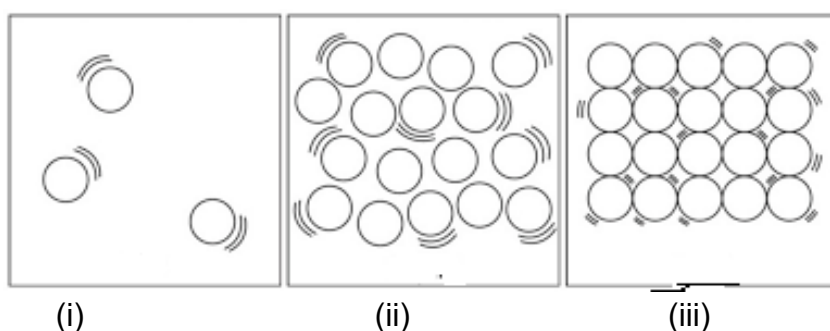
Pick the odd one out and give reason

1. Air, Oxygen, Milk, Nitrogen
2. Strong attraction, Close packing, High compressibility, Fixed volume

Question 2 :

Answer the following :

1. Observe the picture carefully and identify the state of matter :



2. Observe the picture carefully and identify the property exhibited :



3. Rohan and Sia were playing with balloons. Rohan blew air into a balloon and tied it. Sia noticed that the balloon became bigger and rounder. Rohan then took two identical balloons, filled them with air, and tied them to the ends of a wooden stick. He balanced the stick on a string. When he pricked one balloon with a needle, the stick tilted towards the balloon that was still full of air.
- Based on the observation that the balloon became bigger, what property of matter is demonstrated?
 - Why did the stick tilt towards the inflated balloon after one was pricked?
 - Air is a mixture of gases. How are the particles arranged in a gas?
4. Meera took a glass full of water up to the brim. She carefully added two spoons of sugar to it and stirred it gently. She was surprised to see that the sugar disappeared, and even more surprised that the water did **not** overflow from the glass, even though the sugar was added.
- Why did the water level not rise (overflow)?
 - Which state of matter has the maximum intermolecular space?
5. **Scenario:** A sponge is a solid, yet we can easily compress it by squeezing it in our hand.
Question: If solids have a fixed shape and are generally incompressible, why can a sponge be compressed?
6. **Scenario:** A diver is able to cut through water in a swimming pool quite easily.
Question: Why can a diver move through water easily but cannot "cut through" a wooden wall?

The Vikasa school, Tuticorin
Question bank

Subject: Physics

I Semester

Class: 6

Chapter 2: Physical quantities and measurement

Tr Int : RB

I. Multiple choice questions

1. 10 mm is equal to :
a. 1 cm b. 1 m c. 10 dm d. 10 cm
2. The amount of surface occupied by an object is called its :
a. volume b. area c. mass d. length
3. A meter rule is graduated in :
a. m b. cm c. mm d. km
4. A thermometer is graduated in :
a. kelvin b. °C c. g d. cm
5. Out of the following which one unit does not belong to S.I.
a. meter b. pound c. second d. kilogram
6. 1 acre is equal to :
a. 0.09290 m² b. 0.836 m² c. 4046.856 m² d. none of these

II. Assertion and Reason questions

In the questions given below two statements are given –one labeled **Assertion (A)** and the other labeled **Reason(R)**. Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below

- (a) Both A and R are true, and R is the correct explanation of the assertion
(b) Both A and R are true, but R is not the correct explanation of the assertion
(c) A is true, but R is false
(d) A is false and R is true

1. **Assertion (A):** Stopwatch can measure intervals shorter than one second.

Reason (R): Stopwatch counts second using the standard unit of time.

2. **Assertion (A):** A beam balance is used to measure mass.

Reason (R): It compares the unknown mass with a known standard mass.

3. **Assertion (A):** Kilometer is used to measure distances between cities, while centimeters is used to measure the length of a pencil.

Reason (R): Kilometer is an ancient system of measurement, while centimeters is a standard unit of measurement

4. **Assertion(A):** A beam balance is used to measure the quantity of matter in an object.

Reason(R): When the pointer of beam balance is in the centre, it means that the pans are of equal mass.

III. Pick the odd one out and give reason

1. mass, weight, sadness and height
2. Kilometre, hectometre, decametre and centimeter
3. Ruler, parallax error, measuring tape and clock

IV. Case study

Read the given information carefully and answer the questions that follow.

Aryan decided to help his mother set up a small vegetable patch in their backyard. They marked out a rectangular section that was 10 m long and 6 m wide. Aryan wanted to put a simple wire fence around it to keep stray animals away, so he measured all sides carefully. Later, he wanted to calculate the area of the patch. He used a sheet of graph paper to draw it, but he forgot to check the scale of the graph.

In the evening, he accompanied his mother to buy wire and pulses from a local shop. The shopkeeper used a balance with standard metal weights in grams and kilograms.

Answer the following :

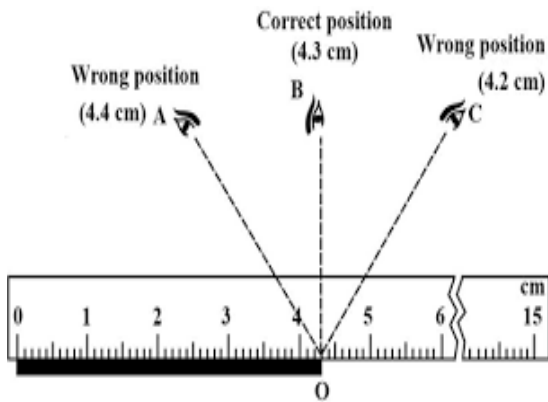
- a. How much fencing wire does Aryan need for the vegetables patch? Which unit should the shopkeeper use to measure it?
- b. Which unit is best suited to express the area of the patch?
- c. What error did Aryan make while using the graph paper?
- d. If the shopkeeper has weights of 50 g, 100 g, 500 g, 1 kg and 2 kg, which weights should he use if Aryan wanted to buy 250 g of lentils?

V. Application based questions

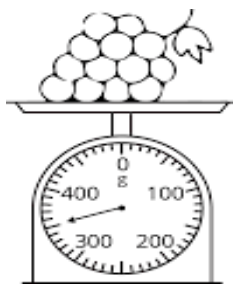
1. Name the instrument which is commonly used to measure the mass of a body. State how it is used.
2. What kind of balance is used for finding the mass of (a) gold ornaments (b) vegetables
3. Why is a clinical thermometer not sterilised in boiling water?
4. Why were standard units of measurement developed? List some SI units of measurement?
5. Explain in detail, the correct way of using a clinical thermometer and a laboratory thermometer.
6. The correct position of eye is important to take measurement. Give reason.

VI. Diagram based questions

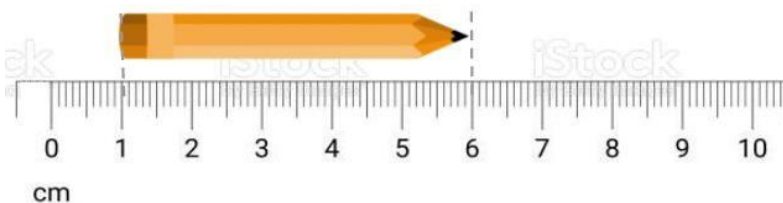
1. What is depicted in the below picture?



2. What is the mass of grapes in kilograms?



3. Find length of pencil in cm.



VII. Numericals

1. Convert the following quantities as indicated :

- 12 inch = _____ ft
- 20 cm = _____ m
- 0.2 km = _____ m
- 1 tt = _____ cm
- 0.2 km = _____ m
- 0.2 cm = _____ mm

2. Express in second (i) 3 minute 15 second
(ii) 5 hour 2 minute 5 second

3. Convert the following quantities as indicated :

- a. 2500 kg = _____ metric tone
- b. 150 kg = _____ quintal
- c. 10 lb = _____ g
- d. 2500 g = _____ kg

4. Convert 24 h time to 12 h time:

- a. 18 : 57
- b. 23 : 25
- c. 09 : 30

5. Convert following 12 h to 24 h time:

- a. 3 : 45 pm
- b. 8 : 20 pm
- c. 11 : 30 am

I. Multiple choice questions

1. What is the SI unit of force?
a. kilogram force b. newton c. gram force d. metre
2. What happens when you apply force in the same direction as motion?
a. The object stops moving c. The object slows down
b. The object moves faster d. The object changes shape
3. The electrical force is a :
a. contact force b. Non-contact c. consequential force d. none of these
4. The force of gravitation causes :
a. only attraction b. only repulsion c. both (a) and (b) d. none of these
5. Which type of force causes a soccer ball to stop rolling on the ground?
a. Gravitational force c. Frictional force
b. Magnetic force d. Electrostatic force
6. Which of the following is/are incorrect about the force of friction?
a. It acts in the direction of motion c. It acts opposite to the direction of motion
b. It depends on the weight of the object d. both (b) and (c)
7. When you apply mustard oil to your hands, which of the following objects will **not** be difficult to hold?
a. Ceramic cup b. Steel cup c. Earthen cup d. Glass cup
8. Which of the following effects of force changes the speed of a moving object?
a. Stopping a moving bicycle c. Changing the shape of a sponge
b. Steering a bicycle to turn d. Pulling a door to open it
9. Deep grooves in the tyres of a tractor :
a. decrease friction b. increase friction c. make it stable d. make it attractive
10. Which of the following is correct?
a. static friction < sliding friction < rolling friction
b. static friction > sliding friction < rolling friction
c. static friction < sliding friction > rolling friction
d. static friction > sliding friction > rolling friction

II. Assertion and Reason questions

In the questions given below two statements are given –one labeled **Assertion (A)** and the other labeled **Reason(R)**. Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below

- (a) Both A and R are true, and R is the correct explanation of the assertion
- (b) Both A and R are true, but R is not the correct explanation of the assertion
- (c) A is true, but R is false
- (d) A is false and R is true

1. **Assertion(A):** Frictional force opposes motion.

Reason(R): Friction is caused by the smooth surfaces of objects.

2. **Assertion(A):** Most of the machines use ball bearing in order to reduce friction.

Reason(R): Ball bearing reduces the area of contact between the surfaces.

3. **Assertion(A):** Sliding friction in machines is replaced by rolling friction .

Reason(R): Sliding friction is always less than rolling friction which prevents wear and tear of machine parts.

4. **Assertion(A):** Frictional force experienced by a 4 kg object will be more in comparison to a 1 kg object, on a hard surface.

Reason(R): The force of friction is directly proportional to the weight of an object on the surface.

III. Pick the odd one out and give reason

1. Gravitational force, frictional force, electrical force, magnetic force
2. Tension force, spring force, gravitational force, normal force
3. Change of direction, change of speed, change of mass, change of shape

Case study

Read the given information carefully and answer the questions that follow.

Rohit was coming back from school when he saw a few people trying to move a heavy metal box. Initially, they found it difficult to move the box. After placing a few cylindrical wooden sticks beneath the metal box, the box moved easily.

1. Why was it difficult to move the metal box earlier?
2. How did the cylindrical sticks help to move the metal box easily?
3. Identify the type of friction in the given situation.

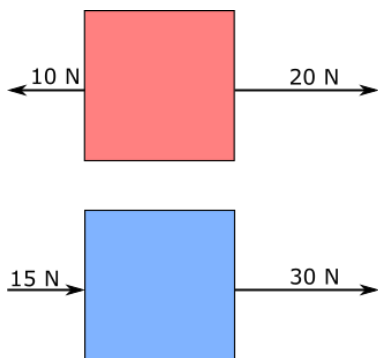
Application based questions

1. Why is it easier to move a suitcase with wheels than one without wheels?
2. Which forces did a rocket like Chandrayaan have to overcome to launch successfully?
3. State the effect produced by a force in the following cases
 - a. when a rubber band is stretched
 - b. when a child pushed a moving swing
 - c. when a girl kicks a ball lying on the ground
 - d. when a cyclist applies the brakes
 - e. when a spring is compressed
4. When you are riding a bicycle on a muddy path, you need to pedal harder than on a smooth road. Why do you think this happens?

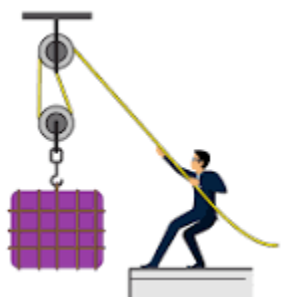
5. If there were no friction on Earth, how would your daily life change?
6. Why do we apply oil to the hinges of doors?
7. Why do the rims of your bicycle's wheel become hot when you apply the brakes?
8. Give three situations that prove that friction is a necessary evil.
9. Why is powder sprinkled on a carom board?
10. Why does a sharp arrow travel farther than a blunt one?
11. In a tug-of-war match, the two teams exert forces of 30 kgf each. What is the net force acting on the rope?

Diagram based questions

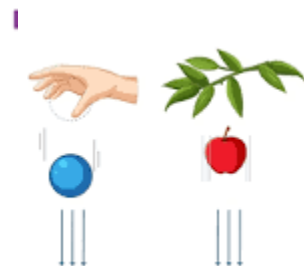
1. Find the resultant force in each of the following figure and the direction in which it acts.



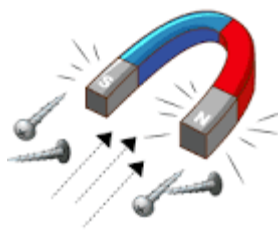
2. Name the kind of forces (contact or non-contact) involved in the following picture.



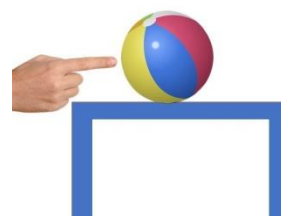
(a)



(b)



(c)



(d)